

Demonstrating poimo as Inflatable, Inclusive Mobility Devices with a Soft Input Interface

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ABSTRACT

In this demo, we showcase poimo, a series of PORTable and Inflatable MObility devices made of a balloon-like soft material called drop-stitch fabric. Poimo has two merits: (1) **inflatability** of a soft and lightweight body allowing users to deflate and carry it on a small scale and (2) **inclusiveness** enabling made-to-order design and fabrication of mobility devices for each user. As an alternative way to conventional rigid controllers, we also introduce a soft and deformable input interface to navigate a sofa-type poimo. Finally, we report two types of field studies: public users riding on two types of poimos on a flat paved road, and the authors riding on a sofa-type poimo in a traditional European environment with inclined stone-paved streets.

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1 INTRODUCTION

New vehicles such as electronic scooters and electronic skateboards are gaining popularity around the world. They are called personal mobility devices, but we consider that they are not yet personalized enough. Current vehicles are bulky and heavy to carry on public transportation, and they cannot be fully customized depending on the preferences and physical characteristics of the user.

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In response to the abovementioned background, we have proposed a new family of personal mobility devices called poimo, considering inflatability and inclusiveness. Inflatable structures provide a lightweight body and allow users to carry the deflated poimo folded up in their backpacks [Sato et al. 2020]. The inclusiveness of poimo is achieved through software that allows users to create one-of-a-kind vehicles based on their physical abilities and body frame, rather than adapting to a standardized product [Niiyama et al. 2020]. Poimo is a series of electric devices with the form of scooter or sportbike, as well as wheelchair and sofa. Steering and wheels can also be made with inflatable structures. Built upon these prior results, we newly present a soft controller based on a sensing technology using conductive foam [Watanabe et al. 2021] and report on two field tests by novice users or on realistic roadways.

2 POIMO

Inflatable Structure. The double-wall drop-stitch fabric is used as body material. Unlike film balloons or vinyl floats, drop-stitch fabric inflatables have a strong, lightweight board-like structure. The hinge, which serves as a steering axis, is also made simply by putting a seam in a single inflatable. We conducted bending tests and confirmed that they can support a human body of about 80 kg.

Inclusiveness. Taking advantage of the unique inflatable structures, we envision a new type of inclusive mobility device that can meet the physical characteristics and preferences of individuals. Thus we have developed an interactive design system (Figure 1C). First, the system provides an initial mobility design considering the skeleton information of the user which was taken by the camera of the user. Then, the user can additionally edit the shape and the color of the initial design. To demonstrate the custom-design capability, we designed several 3D models using our system and built physical samples for two basic templates: a motorcycle and a wheelchair (Figure 1D). Finally, an online user survey was conducted to examine the usability of the design system and to share lessons learned for further improvement of poimo's design and production.

Deformable Input Interface. As an alternative way to conventional rigid input interfaces, we newly designed a soft interface to control poimo using foamin [Watanabe et al. 2021]. foamin is a soft sensor made of conductive foam and a single wire that enables



Figure 1: Poimo as inflatable and inclusive mobility devices. A,B: Poimo is made of drop-stitch fabric that is folded and carried in a small scale, and then inflated to allow users to ride. C,D: Users can design and fabricate their own poimo by considering their body characteristics and preferences on the GUI system.

multimodal gesture detection. For the field study, we created an arch-shaped cushion controller that is operated by the pilot with both hands. Soft sensors made of conductive foam are embedded at both ends of the arch to detect the pilot's gripping gestures and enable the poimo to move forward, turn, and stop. It is also designed to operate not only by gripping with hands, but potentially as a neck pillow that can manipulate poimo by rotating the neck.

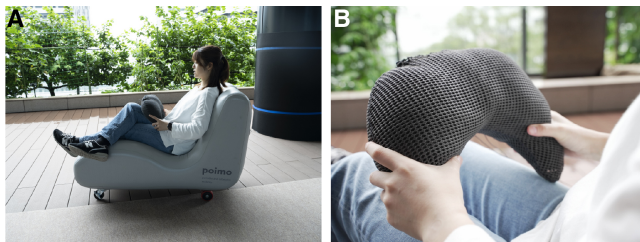


Figure 2: The deformable input interface that allows users to squeeze both grips of the controller to go straight and turn at different speeds depending on the gripping force.

Poimo in the Wild. To advance the social implementation of poimo, we conducted two types of field studies. First, we asked novice users, from children to a pregnant woman, to test-ride the e-scooter type and the sofa-type poimo at the Tokyo Port City and Kashima Stadium, Japan. Approximately 200 participants experienced the ride on poimo (Figure 4A), which demonstrated that poimo can be quickly operated by the public users. Next, to stably ride in the real environment, we designed a new sofa-type poimo. It has five chambers, wheels with motors, battery, and a joystick-based input device for operation, as shown in Figure 3. Using this, we conducted a driving experiment in HUIS TEN BOSCH, Japan, a theme park with a rough stone paving. In this experiment, the authors rode on the poimo for 48 hours and confirmed that it can drive comfortably even on traditional uneven roadways (Figure 4B).



Figure 3: Design of a new sofa-type poimo. The exploded view (left) and the assembled views (center, right).



Figure 4: Field tests done with two sofa-type poimo, with novice users (A) and on uneven stone paving (B).

3 CONCLUSION

We demonstrated the features of poimo, a personal mobility device made possible by inflatable structures. This mobility device has two advantages: (1) inflatability with its soft and lightweight body, and (2) inclusiveness that allows users to design custom-made mobility devices using our interactive design system. In addition, we introduced a deformable input interface for a sofa-type poimo as an alternative to conventional controllers. Finally, we reported on two field studies: (A) a flat paved road experiment in which approximately 200 public users operated the scooter-type and sofa-type poimo, and (B) an inclined rough stone pavement experiment in which the author drove the new sofa-type poimo.

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